Vector Spaces: a Compact Lecture

A fully synthetic demonstration page

Definition

A vector space is a collection of objects that can be added and scaled without leaving the collection.

Example

SOURCE ARTIFACT

Pairs (x, y) form a vector space: add coordinates componentwise and multiply both coordinates by the same scalar.

$$a(u + v) = au + av$$

Figure

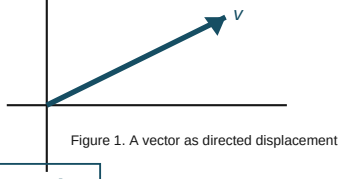

Figure 1. A vector as directed displacement

Result

Two non-parallel vectors in the plane provide a basis for every planar vector.

Proof

Place the two vectors as columns of a 2 by 2 matrix. Non-parallel columns give a nonzero determinant, so the coordinate system has a unique solution.

$$det[u\ \ v] \neq 0$$

Exercise

Decide whether u = (1, 1) and v = (2, -1) form a basis, then express w = (5, 1) in that basis.